



Which Way?

Quick facilitation notes & answer key

Goal

Children choose the single arrow (➡ or ⬅) that moves Bit toward the star. Selecting the correct direction toward a goal is the first planning decision in building an algorithm.

Kindergarten frame (Ontario)

Problem Solving and Innovating — children use directional language and choose a direction toward a goal.

No reading required of the child: read each prompt aloud; the child circles ➡ or ⬅.

Materials

The printed worksheet + a pencil. Optional: floor boxes with Bit and a star to point the way by hand.

Run it (about 15 min)

1. MODEL: 'The star is on box 5, Bit is on box 3. Box 5 is to the right, so Bit goes ➡.'
2. GUIDED: Exercise 1 — star to the right → ➡.
3. YOU TRY: children pick ⬅ when the star is to the left (Ex 2 and Ex 3).
4. Connect: 'A coder first asks: which way do I need to go?'

Answer key (modelled by the run-gate)

Model — Bit 3, star 5 → ➡ (right).

Ex 1 — Bit 2, star 5 → ➡ (right).

Ex 2 — Bit 4, star 2 → ⬅ (left).

Ex 3 — Bit 5, star 1 → ⬅ (left).

Watch for & differentiate

Common slip: always choosing ➡ — have them point from Bit to the star first to feel the direction.

Simplify: use only box positions next to each other (Bit on 2, star on 3).

Extend: ask how many steps in that direction reach the star.

Movement option: stand a child as Bit and place a star; the class calls 'left' or 'right'.

Success looks like

The child circles the arrow pointing toward the star in every case, including when the star is to the left.